

# TITLE PAGE

**Problem Statement:** AI-Powered Behavioral Anomaly Detection for Cybersecurity

**Theme:** Cybersecurity / Industrial Security

**PS Category:** Software

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## ARGUS

Adaptive Risk & Graph-based Unified Security

**Precision 92.13% • Recall 100% • FP Rate 0.32%**

Live: <https://arguskavachh.streamlit.app/>

# ARGUS: Adaptive Risk & Graph-based Unified Security

## PROPOSED SOLUTION

A UEBA-style system that learns per-entity & peer-group 'normal' behaviour from access logs and flags real-time deviations across 7 attack types — no signature rules. Fuses a sequence Transformer, a graph neural network, an Isolation Forest, deterministic hard rules, and a learned calibration layer.

## HOW IT ADDRESSES THE PROBLEM

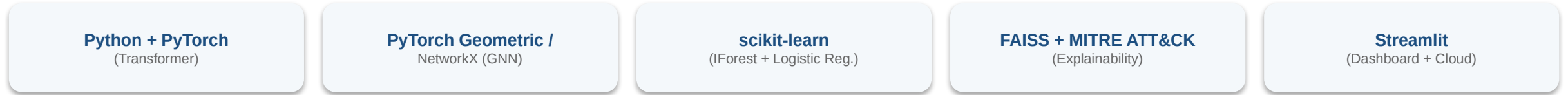
Sequential data → Transformer models session-order behaviour  
Class imbalance → hard rules independent of skewed model confidence  
Concept drift → dedicated monitor separates org change from attack  
Explainability → alerts cite MITRE ATT&CK technique + behavioural deltas  
Cold-start → peer-group baseline + Isolation Forest fallback  
Attack classification → labeled by type, not just flagged (93% accuracy)

## INNOVATION & UNIQUENESS

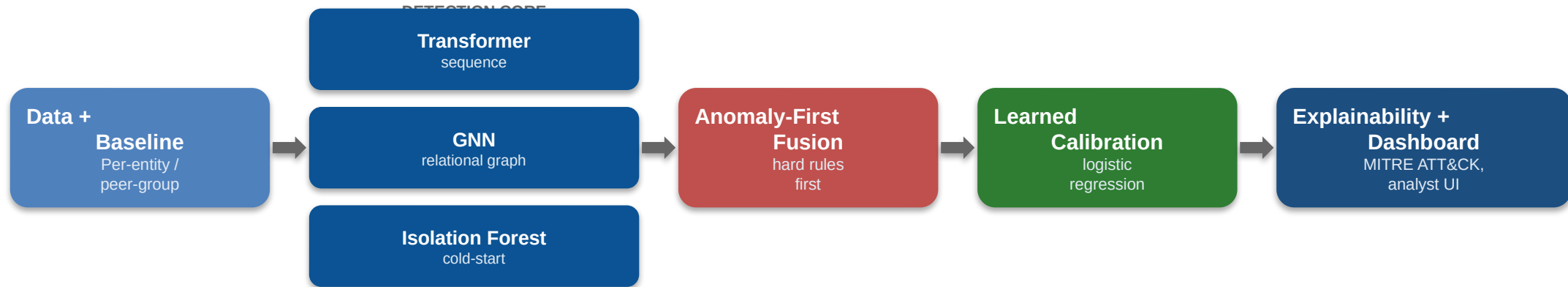
- Anomaly-first fusion: hard rules checked BEFORE model confidence
- 2 strong-looking models independently REJECTED after SHAP + calibration checks found shortcuts, not real signal
- Interpretable 3-coefficient calibration lifted precision 0.745 → 0.921 at unchanged 100% recall
- Architecture extensible to Honeywell Forge Cybersecurity+ / OT telemetry

# TECHNICAL APPROACH

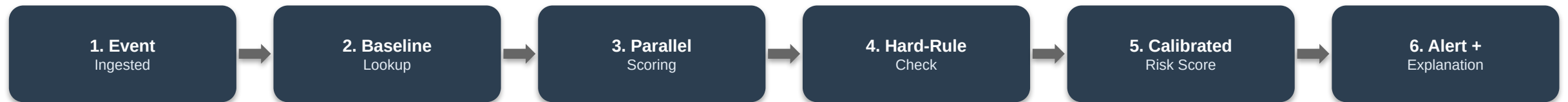
## TECHNOLOGIES USED



## SYSTEM ARCHITECTURE



## PROCESS FLOW — single event, end to end



# FEASIBILITY AND VIABILITY

## FEASIBILITY

Fully working, deployed prototype — not just a concept (live link on Slide 6)

Verified on held-out test data: 92.13% precision, 100% recall, 0.32% FP rate across all 7 attack types

Built end-to-end (data → models → dashboard) within the hackathon window using a documented synthetic dataset

**CHALLENGE:** Rare attack classes have very few test samples

**STRATEGY:** Deterministic hard rules give robust detection independent of model confidence for these classes

**CHALLENGE:** Models can find shortcuts instead of real signals

**STRATEGY:** Verification discipline: SHAP attribution + calibration checks before adopting any result — 2 strong-looking models were caught and rejected

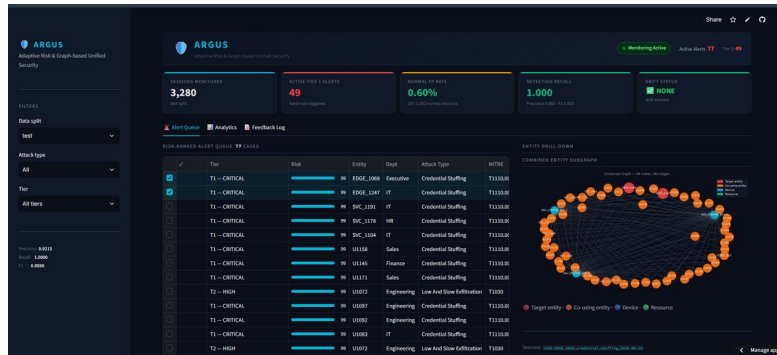
**CHALLENGE:** Scaling to real enterprise / OT data

**STRATEGY:** Modular fusion layer; architecture designed to extend to OT/ICS telemetry (Honeywell Forge alignment)

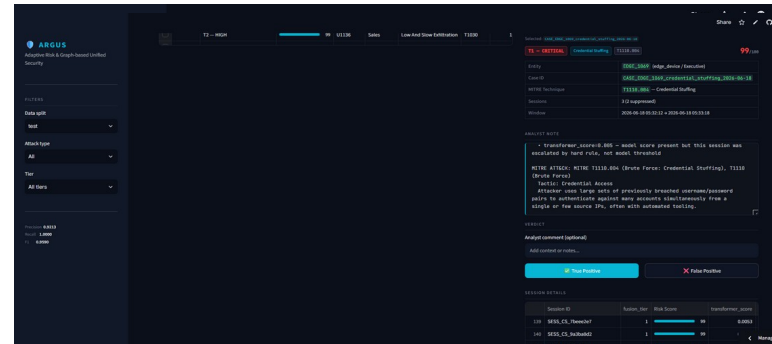
**CHALLENGE:** Legitimate behaviour drifts over time

**STRATEGY:** Dedicated drift monitor distinguishes 'organisation changed' from 'under attack'

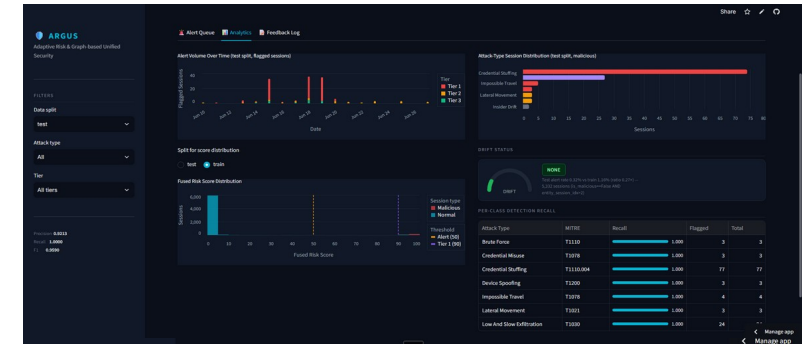
# ARTIFACTS



Risk-ranked alert queue + entity subgraph



Explainability: MITRE ATT&CK-grounded analyst note



Analytics: score distribution & per-class recall

```
if hard_rule_fired(session):  
    tier = 1  
else:  
    score = logistic_calibration(transformer_score, if_norm, graph_boost)
```

Core logic — `anomaly_first_fusion.py`

- # checked BEFORE model confidence
- # geo-velocity / fp-mismatch / IP fan-in

## Full source & solution proposal

GitHub: [github.com/atharv1945/Argus](https://github.com/atharv1945/Argus)

Live app: [arguskavachh.streamlit.app](https://arguskavachh.streamlit.app)

# RESEARCH AND REFERENCES

**Live Dashboard :** <https://arguskavachh.streamlit.app/>

**Source Code (GitHub) :** [github.com/atharv1945/Argus](https://github.com/atharv1945/Argus)

**MITRE ATT&CK Framework :** [attack.mitre.org](https://attack.mitre.org) — technique taxonomy grounding the explainability layer

**Honeywell Forge Cybersecurity+ / Cyber Proactive Defense :** Real-world OT/ICS behavioural-baselining products this architecture is designed to extend towards

**UEBA industry background :** Reviewed alongside commercial UEBA approaches (e.g. Exabeam, Splunk, Microsoft Sentinel) to ground the problem space